



Mussel-inspired anti-biofouling and robust hybrid nanocomposite hydrogel for uranium extraction from seawater



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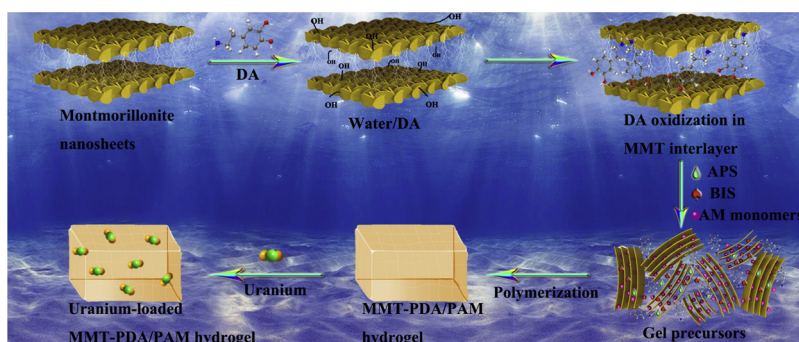
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GRAPHICAL ABSTRACT



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ABSTRACT

A major challenge of uranium extraction from seawater (UES) is to effectively block the biofouling without destroying the ecological balance, especially prevent the attachment of macroalgae on the surface of the adsorbent. Herein, a robust montmorillonite-polydopamine/polyacrylamide nanocomposite hydrogel is reported by a two-step method, including PDA intercalation MMT and further free radical polymerization with AM monomers. The interpenetrating structure of hydrogel lead to high water permeability with the swelling ratio of 51, which could fully facilitate the internal accessible sites exposure and increase the uranium diffusion. As a result, a high adsorption capacity of 44 mg g^{-1} was achieved in lab-scale dynamic adsorption. Most importantly, the prepared anti-biofouling hydrogel adsorbents display excellent anti-adhesion ability towards *Nitzschia* after 8 days contact. The adsorption capacity of uranium can reach $2130 \mu\text{g g}^{-1}$ in algae-contained simulated seawater. This hydrogel also exhibited a long service life of acceptable mechanical strength and adsorption capacity after at least 6 adsorption-desorption cycles. This new anti-biofouling nanocomposite hydrogel shows great potential as a new generation adsorbent for UES.

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1. Introduction

Uranium is one of the most important fuel for nuclear reactors among all radionuclides, so the exploitation of uranium is beneficial for the development of energy industry (Wu et al., 2017). The world's oceans, where uranium is found evenly at a concentration of $3.3 \mu\text{g L}^{-1}$ with the total amount about 4.5 billion tons (Ma et al., 2015), providing an alternative source of uranium from conventional terrestrial mining for nuclear fuel. Adsorption is a facile and cost-effectiveness way for uranium extraction from seawater (UES) of a matter owing to the convenience in design of the adsorbents. Lately, the various uranium adsorbent materials, including activated carbon and its derivatives (Coleman et al., 2003; Starvin and Rao, 2004), porous organic polymers (Sun et al., 2018a; Wang et al., 2019), modified mesoporous silica (Gunathilake et al., 2015; Li et al., 2017a) and polyphenolic compounds (Liu et al., 2013), are widely investigated. However, it is still a challenge for these reported adsorbents to conquer the biofouling during the process of UES.

Biofouling caused by marine living creature severely limits the adsorption capacity and decreases the reusability of the adsorbent (Abney et al., 2017; Zhang et al., 2016; Yu et al., 2019; Yang et al., 2019). To cope with this ubiquitous problems, many research have been devoted to develop the antifouling adsorbents. Cheng et al revealed that the introduce of nano-TiO₂ into amidoxime functionalized wool fiber can effectively impart antibacterial activity toward staphylococcus aureus and escherichia coli, and the adsorption amount of Wool-AO@TiO₂ was 113.12 mg g^{-1} (Wen et al., 2018). Hua et al prepared amidoxime and guanidine co-functionalized polypropylene nonwoven fabric (PP-g-AO) for potential UES, which can efficiently killing the bacteria after 20 h culture with the adsorption capacity of 112 mg g^{-1} (Zhang et al., 2018a). Our group also investigated the antifouling adsorbents for UES, which revealed that introducing substances with antibacterial ability (such as: chitosan and Ag nanoparticles) can greatly inhibit the growth of diatom (Guo et al., 2018; Zhang et al., 2019a). Although these literatures have reported adsorbents with anti-biofouling activity, but they also cause environmental ecological balance damage for killing algae. Therefore, it is significant to develop an eco-friendly uranium adsorbent with anti-adhesion property. To the best of our knowledge, there has been no reports on this.

The hydrophilic polymers exhibit less marine microorganisms adhesion, which can be applied for long-term marine antifouling (Yang et al., 2012; Chen et al., 2015). Hydrogels, a well-defined kind of three-dimensional (3D) networks, formed by hydrophilic polymer chains inserted in an aqueous environment. The unique interpenetrating networks with numerous chelating groups, which makes hydrogel having stronger affinity and more adsorption sites (Pan et al., 2016; Zhang et al., 2017; Ma et al., 2017; Zhang and Khademhosseini, 2017). However, hydrogels are too brittle due to the high water content in the polymer networks (Zhang and Khademhosseini, 2017), which is difficult to response the shock of ocean currents in the actual marine environment. As a solution, hybridization hydrogels formed with nanomaterials via blending multiple components (Li et al., 2014a; Sun et al., 2012) or crystallite cross-linking (Li et al., 2014b), may possess substantially improved mechanical property. Montmorillonite (MMT) nanosheet, because of their highly tunable surface properties and strong mechanics, contribute to enhanced mechanical performances of bulk hydrogels through interacting with polymer chains (Annabi et al., 2016; Gaharwar et al., 2014; Haraguchi and Takehisa, 2002). Nevertheless, hybridization hydrogels confront another problem, namely, the poor interface compatibility between the nano-sized additive and the polymer matrix, leading to unsatisfied stability and nonuniform dispersion of the hydrogel.

Inspired by the adhesive property of mussel foot proteins to solid surfaces, polydopamine (PDA) surface chemistry modification has arisen as a popular strategy in the materials science (Liu et al., 2014). Numerous studies have shown that PDA was used to enhance interfacial

interactions between polymer matrices and nanomaterials (Jiang et al., 2014; Yang et al., 2015). Another valuable feature of PDA lies in its chemical structure that incorporates many functional groups such as catechol, amine, and imine. These accessible active sites can not only serve as the anchors for the uranium but also improve the hydrophilic of the hybrid materials, which is beneficial to the UES and anti-biofouling. Meanwhile, as the naturally occurring melanin, PDA processes superb biocompatibility, which will not destroy the ecological balance of the ocean (Liu et al., 2014).

Herein, a robust montmorillonite-polydopamine/polyacrylamide (MMT-PDA/PAM) nanocomposite hydrogel was fabricated through two-step process, involving the intercalation of PDA into MMT nanosheets and further free radical polymerization with AM monomers. The PDA-intercalated MMT can effectively solve the problem of interface compatibility between the MMT nanosheet and PAM matrix as well as improved the mechanical behavior and stability of the hydrogel. The effect of pH, concentration, temperature, contact time, coexisting ions, ionic strength, and algae coexistence environment on the uranium uptake was investigated. Due to the high hydrophilic and the various functional groups, the fabricated adsorbent showed impressive anti-adhesion activity and enhanced uranium extraction capacity. As far as we know, this study is the first to propose an eco-friendly uranium adsorbent that prevents algae from adhering to the surface of the adsorbent without compromising ecological balance, thereby achieving the purpose of anti-biofouling.

2. Experimental

2.1. Synthesis of MMT-PDA/PAM hydrogel

The intercalation and restricted oxidation of DA between MMT nano-layers were processed by dispersing 0.6 wt.% DA in 80 wt.% deionized (DI) water to form DA aqueous solution, then the 10 wt.% MMT was added into the DA solution to form the MMT-DA suspension. The suspension was stirred vigorously for 6 h to obtain MMT-PDA mixture. Hydrogels were prepared by the free radical polymerization using solutions consisting of monomers AM, initiator APS and in the vicinity of a crosslinking agent BIS. 2.5 g AM, 20 μL TMEDA, 0.2 wt.% MBA and 15 wt.% APS were added to the MMT-PDA suspension under stirring in an ice water bath. After 20 min, the ice water bath and stirrer were removed and the mixed suspension was polymerized automatically to prepare MMT-PDA/PAM hydrogel (Fig. 1). The different content of MMT and DA of the hydrogels are listed in Tables S1 and S2. Then, by comparing the amount of adsorption to determine the optimal ratio. As can be seen in Fig. S1, the MMT₁₀-PDA_{0.6}/PAM has the maximum adsorption capacity, so 0.6 wt.% DA and 10 wt.% MMT was chosen as the optimal ratio. For comparative study, the MMT₁₀-PDA_{0.6} and MMT₁₀/PAM were prepared according to same procedure, but just without add the AM and DA, respectively.

2.2. Sorption experiments

The solutions with different uranium concentrations were obtained by diluting the stock uranium solution with the proper amount of DI water. The pH levels of the solutions were adjusted by $0.5 \text{ mol L}^{-1} \text{ HNO}_3$ or $0.5 \text{ mol L}^{-1} \text{ Na}_2\text{CO}_3$. The adsorbent dosage was regarded by weighing the dry hydrogel and the gel was swollen by DI water for five hours before adsorption. The adsorption amount (Q_e) and the removal rate of uranium adsorbed per unit mass of the adsorbent was calculated according to the eq S1 and eq S9, respectively. The uranium-loaded adsorbent was eluted and regenerated with $0.05 \text{ mol L}^{-1} \text{ NH}_4(\text{CO}_3)_2$, further washed with DI water. Seawater used in this work was collected from Yellow Sea. All the experiments were conducted thrice under identical condition and the relative errors of the data were within 5%.

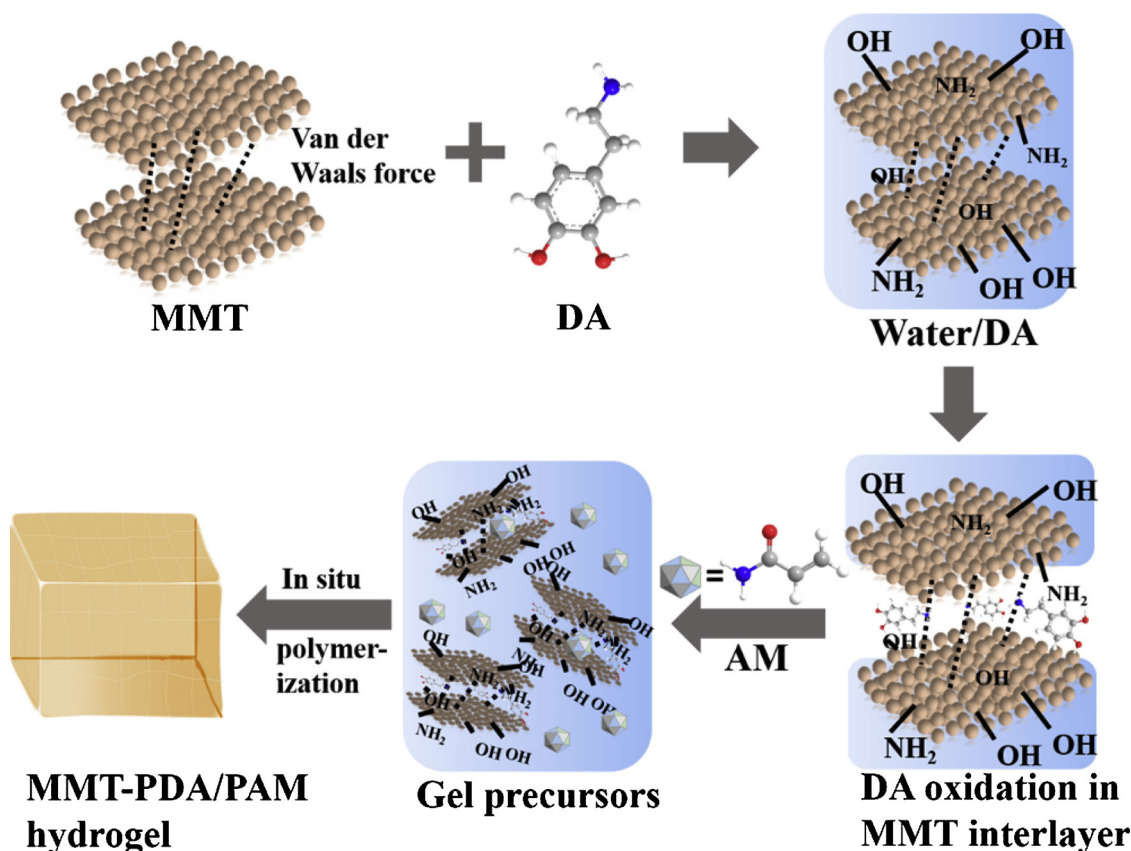


Fig. 1. Synthetic scheme of MMT-PDA/PAM hydrogel.

2.3. Anti-biofouling experiments

As a typical biofouling microorganism, diatom is commonly found in the marine environment, which is easy to adhere to the surface of artificial materials. Therefore, *Nitzschia* (a kind of diatom) was selected as the main research object to evaluate its anti-biofouling ability. Specifically, the *Nitzschia* was cultivated at $21 \pm 2^\circ\text{C}$ in algae culture solution with a 12 h light and 12 h dark (L/D) cycle of fluorescent illumination and stirred every 12 h. 50 mL of *Nitzschia* suspension and 20 mg adsorbents were added into conical flasks. The conical flasks were incubated and kept for further analysis at defined time periods in a biochemical incubator.

2.4. Apparatus

The functional groups were detected using a FTIR Avatar 370 spectrophotometer. XRD patterns were performed using a Rigaku D/max-IIIIB diffractometer with Cu K α irradiation and operating at 40 kV and 150 mA. SEM (JSM-6480A, Japan Electronics) was used to observe the inner structure morphologies of the MMT-PDA/PAM hydrogel. The surface chemistry of adsorbents before and after adsorption process was determined by XPS (ESCALAB 250Xi). The zeta potential of the samples was measured by nanoZS90 in a series of aqueous with different initial pH. The concentrations of uranium were detected by ICP-AES and ICP-MS. To determine the anti-adhesion properties of the MMT10/PAM and MMT10-PDA0.6/PAM, the experiments were carried out by counting the concentration of *Nitzschia* on a hemocytometer via optical microscope (Leica DFC295) and observing the attachment of *Nitzschia* on the surface of the adsorbents via fluorescence microscope (Nikon DS-Ri2).

3. Results and discussion

3.1. Material characterization

XRD was conducted to demonstrate the insertion of MMT nanosheets during hydrogel formation (Fig. 2(a)). In the XRD profile of MMT, a broad peak at 6.28° was observed, which was assigned to d (001) diffractions (the d(001) spacing: 1.4 nm). After the introduction of the PDA, the d(001) spacing of increased to 1.7 nm, which indicated that the PDA has been inserted into the MMT nanosheets and resulted in an increased height. With the addition of AM, the d(001) spacing further increased to 1.9 and 2.1 nm for MMT₁₀/AM and MMT₁₀-PDA_{0.6}/AM, respectively, which indicated that the further expansion of the MMT nanosheets space. It is worth note that the XRD patterns of MMT₁₀/PAM and MMT₁₀-PDA_{0.6}/PAM hydrogel were featureless after the AM polymerized in situ to PAM. This may be due to in situ polymerization of AM and complete stripping of MMT nanosheets (Chen et al., 2016). The FT-IR of the MMT₁₀-PDA_{0.6}/PAM hydrogel was compared with the pregel solution and showed the appearance of the saturated C-H bonds (2932 cm^{-1}) in the hydrogel sample (Fig. 2(b)), thus indicating the condensation occurrence (Xiao et al., 2017). For the swelling experiment, the water uptakes of the MMT₁₀-PDA_{0.6}/PAM and MMT₁₀/PAM hydrogels were measured by a gravimetric method. The swelling ratio was defined as:

$$\text{Swelling ratio} = \frac{W_s - W_d}{W_d} \quad (1)$$

and the moisture content of the hydrogel was calculated according to:

$$\text{Moisture content} = \frac{W_s - W_d}{W_d} \times 100\% \quad (2)$$

where W_s and W_d are the weights of the fully swollen and dry hydrogel,

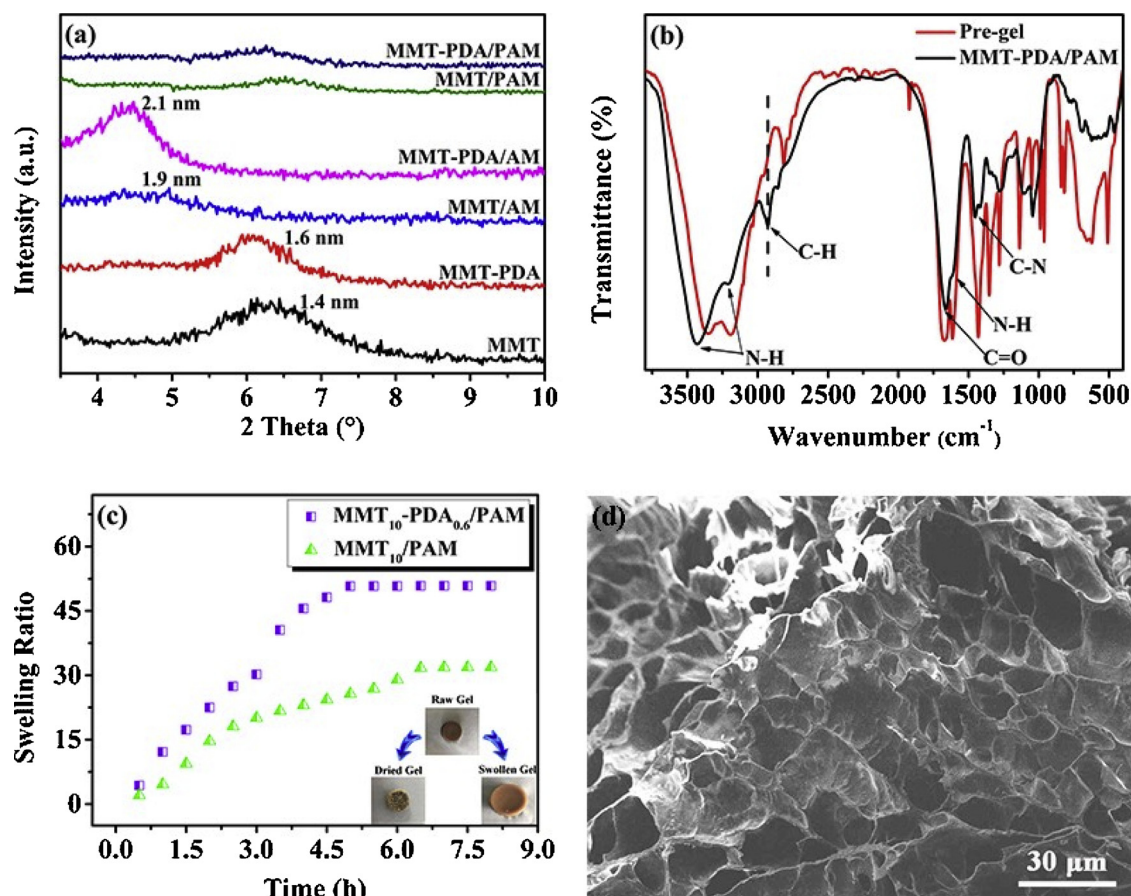


Fig. 2. (a) XRD patterns of different materials, (b) FT-IR spectra of pre-gel solution and MMT₁₀-PDA_{0.6}/PAM hydrogel, (c) Swelling ratio of MMT₁₀/PAM and MMT₁₀-PDA_{0.6}/PAM (insets are the photographs of raw, dried and swollen MMT₁₀-PDA_{0.6}/PAM). (d) SEM image of MMT₁₀-PDA_{0.6}/PAM hydrogel after freeze-dried.

respectively. The dried MMT₁₀-PDA_{0.6}/PAM hydrogel swelled as time going in water until reaching equilibrium at around 5 h with the swelling ratio of 51 (Fig. 2(c)). The hydrogel showed good water retention as the moisture content of the raw and swollen hydrogel was calculated to be 60.90% and 81.45%, respectively. It also has the ability of repeat drying and swelling without rupture (inset, Fig. 2(c)), which are beneficial for its application in UES. The microstructure of the adsorbent was characterized by SEM after freeze-dried (Fig. 2(d)). The hydrogel presents a interconnected macroporous network and highly density of crosslinks, which can provide good mechanical strength and facilitate the diffusion of uranium into the hydrogel (Ma et al., 2016). The 3D network structure is formed by the interaction of MMT, PDA and PAM, because no obvious pore tunnel structure were observed in individual component and two components (Fig. S2(a-d)).

3.2. Mechanical property of the hydrogels

Hydrogel with tough, resilient and stretchable is suitable for UES. To assess the mechanical properties of the MMT₁₀-PDA_{0.6}/PAM, the compressive stress versus strain was firstly investigated. As shown in Fig. 3(a), the MMT₁₀-PDA_{0.6}/PAM hydrogel can tolerate a high compression strain of 90% without cracking. Moreover, the hydrogel can recovered automatically to its original appearance after the compression force was removed. The MMT₁₀-PDA_{0.6}/PAM hydrogel was also subjected to fatigue hysteresis tests with 30 compressive cycles at a large ϵ of 90% (Fig. 3(c)). The recovered MMT₁₀-PDA_{0.6}/PAM hydrogel presented similar shape to the original one at 30th cycle during the whole experiment conditions, highlighting their structural robustness. In addition, the MMT₁₀-PDA_{0.6}/PAM hydrogel can be stretched to 7 times of its original length and then returned to its original length of

stress removal, as shown in Fig. 3(b). Fig. 3(d) displays a representative tensile curves of different hydrogels. The values of tensile strength at break are 52, 103, 41, 18, 31 KPa and the values of elongation at break are 803%, 902%, 1678%, 1348%, 1227% for PAM, MMT₁₀/PAM, MMT₁₀-PDA_{0.6}/PAM, PDA_{0.6}/PAM and seawater soaked MMT₁₀-PDA_{0.6}/PAM hydrogel, respectively. From the above data, the MMT₁₀-PDA_{0.6}/PAM hydrogel exhibited the maximum strain, and its' value only decreased by about 27% even after 30 days of seawater immersion. The excellent mechanical property is beneficial for practical application of hydrogel adsorbents to withstand the turbulence of seawater.

3.3. Uranium sorption studies

The pH of the solution is an essential factor to note during the adsorption process, because the protons not only affect the surface charge of the adsorbent but also affect the speciation of uranium in solution. The adsorption capability towards uranium of three adsorbents were investigated over the pH range from 3.0 to 8.5. As can be seen in Fig. 4(a), the amount of uranium adsorbed on MMT₁₀-PDA_{0.6}/PAM hydrogel (229.08 mg g⁻¹ at pH = 6.5) is remarkably higher than that of uranium on MMT₁₀-PDA_{0.6} and MMT₁₀/PAM. Interestingly, as the pH gradually increased from 6.5 to 8.5, the adsorption capacity of MMT₁₀-PDA_{0.6}/PAM hydrogel just decreased slightly. It is well known that pH = 8.3 is the closest to the pH of real seawater circumstances the adsorption amount can still reach 200.45 mg g⁻¹ at pH = 8.0 and 195.53 mg g⁻¹ at pH = 8.5. This feature makes the MMT₁₀-PDA_{0.6}/PAM hydrogel a candidate for UES.

The variation of adsorption amount is due to the surface charge of adsorbents and the uranium species are highly pH dependent. UO_2^{2+} is the predominant species at pH < 5.5, the poor adsorption amount can

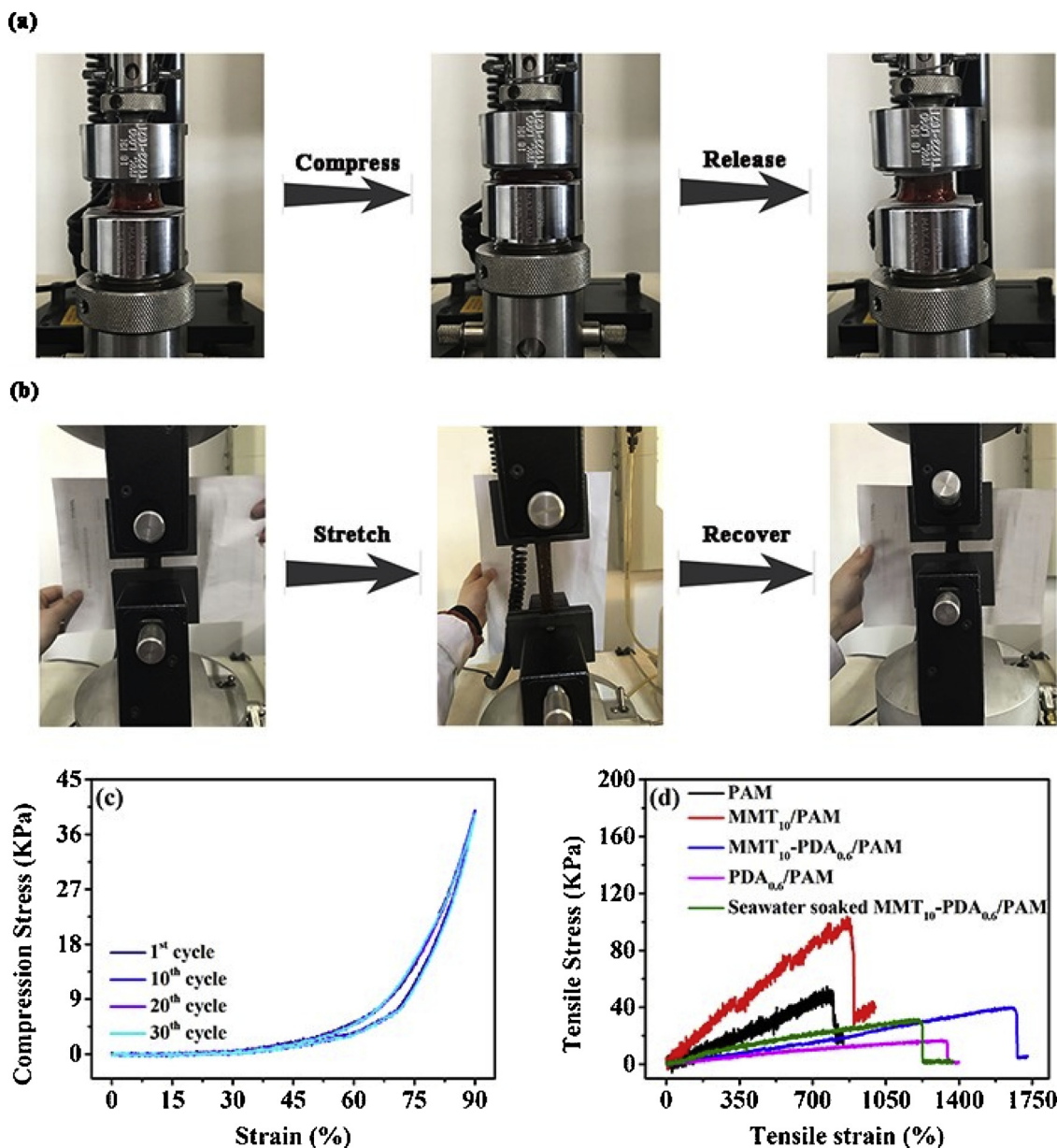


Fig. 3. (a) The compression process, (b) the stretching process, (c) the fatigue compression test with 30 cycles ($\varepsilon = 90\%$) of the MMT₁₀-PDA_{0.6}/PAM hydrogel, (d) representative tensile stress-strain curves of different hydrogels.

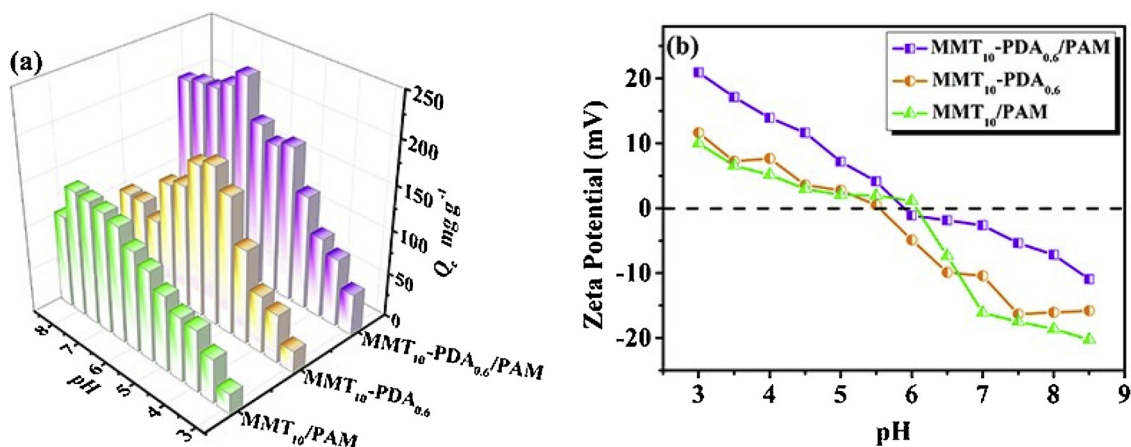


Fig. 4. (a) Effects of pH values on the adsorption capacities of uranium on three adsorbents, (b) zeta-potentials of three adsorbents under different pH values.

be associated with the competition between UO_2^{2+} and H^+ for the accessible active sites and the electrostatic repulsion between UO_2^{2+} and the adsorbent surface (Wu et al., 2017). The zeta potential value became negatively charged with increasing pH to 5.9, which enhanced the electrostatic attraction between sorbents and positively charged uranium species (Fig. 4(b)) (Wang et al., 2012). With pH increasing to 6.5, uranium hydrolyzed to colloidal or oligomeric species, such as $[\text{UO}_2(\text{OH})]^+$, $[\text{UO}_2(\text{OH})_2]^{2+}$ and $[(\text{UO}_2)_3(\text{OH})_5]^+$, which could be adsorbed better than the free hydrated ions (Liu et al., 2019; Wang et al., 2018a). In addition, the content of H^+ gradually reduced, leaving more unoccupied sites for uranium. However, at even high pH value (neutral or alkaline condition), uranium is mainly in the forms of $\text{UO}_2(\text{CO}_3)_2^{2-}$ and $\text{UO}_2(\text{CO}_3)_3^{4-}$ (Zhang et al., 2015; Huang et al., 2018), which reduced the accessibility toward the negatively charged exterior of $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel, leading to slightly weakening adsorption amount of uranium. Therefore, an optimum pH value for the effective adsorption is regarded as 6.5 for the following tests. Solid content is another essential factor affecting the adsorption efficiency of uranium and the economic benefit. The effect of solid content ($0.1\text{--}1.0\text{ g L}^{-1}$) on uranium adsorption was investigated by exploring the optimum amount of adsorbent (Fig. S3). The result showed that 0.4 g L^{-1} was the optimum solid content for the adsorption process.

To assess the adsorption capacity of $\text{MMT}_{10}\text{-PDA}_{0.6}$ (Fig. S4), $\text{MMT}_{10}/\text{PAM}$ (Fig. S5) and $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ (Fig. S6) for uranium, equilibrium values were collected by varying the concentrations of uranium in the range of $10\text{--}500\text{ mg L}^{-1}$ (the detail was in 'Sorption Isotherms' section of supplementary information). The saturation adsorption capacities were determined to be 540.54, 502.21 and 617.28 mg g^{-1} for above three adsorbents, respectively. To describe the sorption isotherms distinctly, the Langmuir, Freundlich and Dubinin-Radushkevich (eq S2-S5) were used. It can be found that all the adsorption isotherms fit the Langmuir model with correlation coefficients higher than 0.98 (Table S3). Remarkably, the adsorption capacity of $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ outperforming most of the previously reported acrylamide-based materials and polydopamine functionalized adsorbents (Table 1). Such an outstanding saturation uranium adsorption capacity can be attributed to the tremendous affinity together with a large number of highly accessible groups. The adsorption process on three adsorbents were also investigated at 25°C , 35°C and 45°C (see Fig. S7). The adsorption amount of uranium increases with rising the temperature, which indicate the high temperature driven adsorption process. The thermodynamic parameters were obtained by the eq S6-S8 and the calculated values were listed in Table S4.

A short operation process is also greatly desirable in conformity with real requirement of UES. As shown in Fig. 5(a), the adsorption kinetics can be comprehend as three-step. In the initial stage, the uranium diffusion from liquid phase to the exterior surface of adsorbent which exhibited as the sharp adsorption. Subsequently, uranium gradually diffusing into the inner surface of adsorbents, the adsorption process tends to be gentle. Finally, with active sites become gradually occupied, repulsive force on solid-liquid interface was enhanced, resulting in a low adsorption rate and then steadily approach to equilibrium. The rapid kinetics may attribute to the synergistic effects arising

from large amount of chelating groups coupled with 3D pores allowing for rapid diffusion of uranium throughout the materials. Three adsorbents reached their equilibrium capacity after 30 min, 90 min and 120 min for $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$, $\text{MMT}_{10}\text{-PDA}_{0.6}$ and $\text{MMT}_{10}/\text{PAM}$, respectively. These data are superior to other reported adsorbents, which generally take longer contact time, typically hours to days (Yuan et al., 2018a; Wang et al., 2018b; Das et al., 2009). For shed light on the adsorption reaction, the data were fitted with three different models (Fig. 5(b-d)). According to the details of kinetic parameters in Table S5, the pseudo-second-order kinetic was the better fitting model due to the higher correlation coefficient ($R^2 = 0.99$), which indicates that chemical adsorption is the rate controlling step for uranium extraction to $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel (Zhang et al., 2019b).

The reactivity of $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel between U and the other interference ions (Sr, Ni, Na, Mg, K, Co, Ca, Ba, V and Fe) were conducted under the same molar ratio. As can be seen from Fig. 6(a), the $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel possesses an excellent selectivity for uranium with a high removal rate of over 85%. The distribution coefficient (K_d) value as used for the determination of the affinity and selectivity of adsorbents for uranium, and is given by the eq S6. The K_d of uranium (nearly 17647) was far above than other interference ions, which further proves the outstanding affinity of $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel for uranium (Fig. 6(b)). Three factors can be lead to the excellent selectivity. Firstly, on the basis of HSAB theory, hard bases (O, N) interact more strongly with hard acids uranium (Pearson, 1966). The nitrogen, oxygen-containing functional group in $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ has strong interaction with U, which can form more stable complexes than other metal cations. Secondly, considering that the interaction ability with functional groups, the f orbital of uranium is easier to combine than the d orbital of other heavy metals (Bai et al., 2015). Thus, the adsorption capacity for U ions is better than Ni, Co, V, Fe ions. Thirdly, the bond length and strength will also affect the adsorption capability. The U-N bond length is significantly shorter than the length of some metal-nitrogen cations (such as Sr, Ni, Na, K, Ca and Ba ions), which indicate that a strong affinity of nitrogen organic groups toward U ions. U-N bond length is longer than other metal-nitrogen cations (such as Mg and Fe ions), leading to a depth effect for the nitrogen-combined with U ions (Wu et al., 2015). Considering the various ionic species in seawater, the effect of ionic strength on U(VI) adsorption was also investigated. The results in Fig. S8 demonstrate a slight decrease of uranium adsorption with increase the concentration of NaCl and NaHCO_3 , indicating the formation of a hydrated layer on the surface of the $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel and the competition of Na^+ ions with U(VI) for the active sites of adsorbents. However, the overall removal rate of uranium was not significantly change at any condition, indicating that the adsorption mechanism would be inner-sphere surface complexation (Husnain et al., 2017). Therefore, the as-obtained results forecast the formation of stable chelates between uranium and the amino-groups on the surface of $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel (Zhang et al., 2018b).

Table 1
Uranium adsorption performance of various adsorbent materials.

Adsorbents	Equilibrium adsorption amount	Contact time	pH	Solid content (g L^{-1})	Ref.
$\text{Fe}_3\text{O}_4/\text{PDA}$ -5	44.5 mg g^{-1}	150 h	5	1.5	(Wu et al., 2017)
$\text{GO}/\text{PDA}/\text{CS}$ aerogel	415.9 mg g^{-1}	15 min	6.0	–	(Liao et al., 2018a)
$\text{RGO}/\text{PDA}/\text{oxime}$	675 mg g^{-1}	6 h	5.0 ± 0.1	–	(Qian et al., 2018)
WPC/PDA	384.6 mg g^{-1}	60 min	7.0	0.5	(Zhu et al., 2018)
$\text{AT}/\text{PDA}/\text{CS}$ aerogel	165.6 mg g^{-1}	40 min	5.0	0.4	(Liao et al., 2018b)
(P(DEA-co-AA)) microgels	$8.548 \times 10^{-4}\text{ mol g}^{-1}$	5 h	4.0 ± 0.05	0.4	(Tonghuan et al., 2013)
Clay biomass complexes (BMMT)	$0.53 \pm 0.07\text{ mmol g}^{-1}$	3 h	5.5	–	(Olivelli et al., 2017)
$\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$	617.28 mg g^{-1}	30 min	6.5	0.4	This work

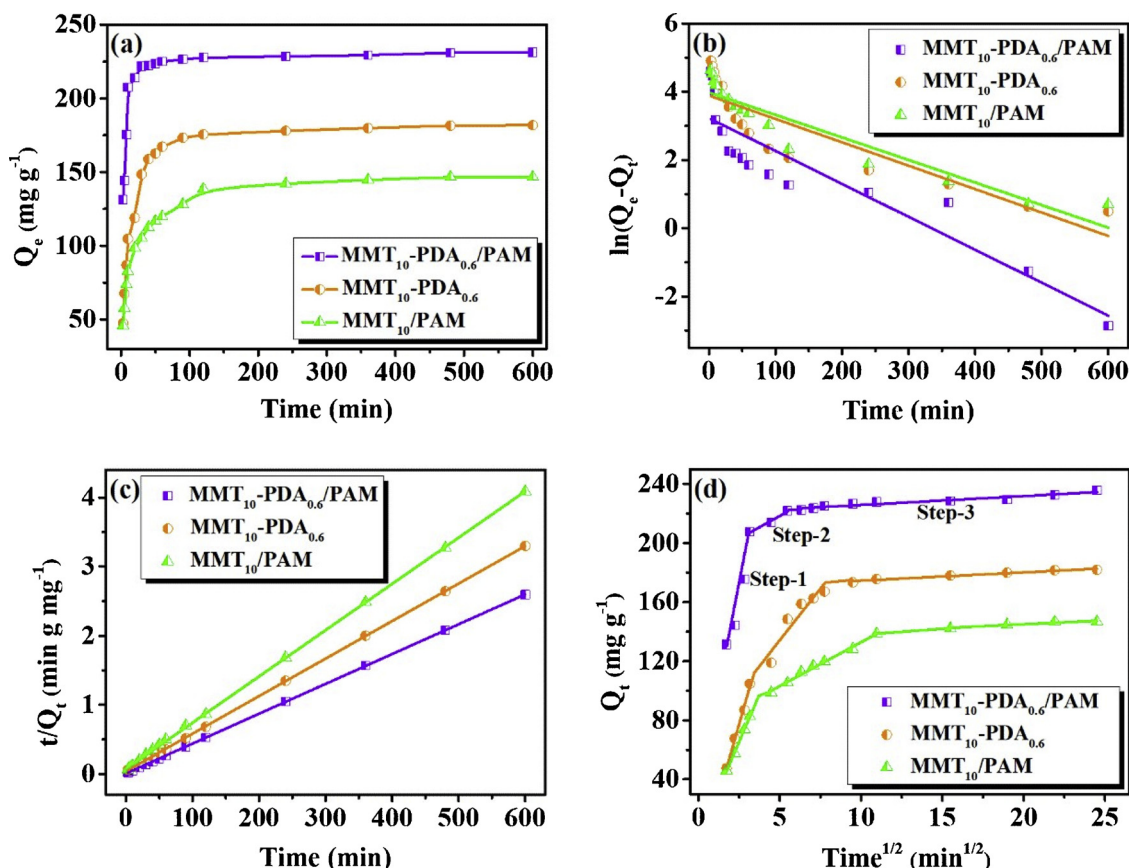


Fig. 5. (a) Effect of contact time on the uranium adsorption by three adsorbents, and corresponding (b) pseudo first-order kinetics, (c) pseudo second-order kinetics, (d) morris-weber model.

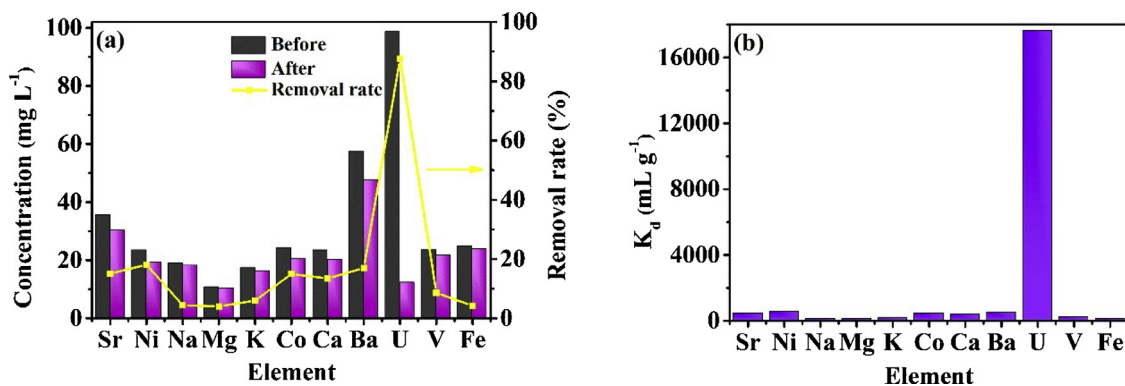


Fig. 6. (a) Effect of interference ions on the adsorption of U by $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$, (b) K_d value of U and other interference ions.

3.4. Uranium extraction from simulated and natural seawater

Given that nuclear power will be decisive for future low carbon energy generation, there is a strong motivation to develop adsorbent materials that can efficiently seize uranium from seawater as well as easy to recovery. In this regard, we initially applied the $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ to test uranium extraction from the simulated seawater, which spiked with $\sim 10.3 \text{ mg L}^{-1}$ of uranium in the presence of excess HCO_3^- and Na^+ (NaHCO_3 0.198 g L^{-1} and NaCl 25.6 g L^{-1}) (Gunathilake et al., 2015), giving rise to a high removal rate of more than 90% (Fig. 7(a), curves) in the whole concentration range (from 3.68 to $998.23 \text{ } \mu\text{g L}^{-1}$). As for $\text{MMT}_{10}/\text{PAM}$, the removal rate is just between 72% and 85%. Thus suggesting $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ hydrogel is a promising candidate for UES. We further comparatively evaluated the adsorption ability of $\text{MMT}_{10}/\text{PAM}$ and $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ toward

uranium in natural seawater and in extra uranium (10 to $1000 \text{ } \mu\text{g L}^{-1}$) spiked natural seawater from Yellow Sea in China. Similar results were obtained, the $\text{MMT}_{10}/\text{PAM}$ showed moderate adsorption of uranium (the removal rate between 45% and 60%, Fig. 7(a), light grey columns), however after the introduction of PDA into hydrogel, $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ showed enhanced removal rate in each test solutions (the removal rate between 70% and 84%, Fig. 7(a), dark grey columns). It is noteworthy that due to the absence of various competing ions and micro-organisms, both $\text{MMT}_{10}/\text{PAM}$ and $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ exhibited obviously increased removal efficiency in simulated seawater over that obtained in natural seawater.

In addition, the adsorption capability toward uranium of the $\text{MMT}_{10}\text{-PDA}_{0.6}/\text{PAM}$ was investigated in a laboratory-scale continuous simulated dynamic adsorption system (the detail was in 'Dynamic Adsorption' section of supplementary information). From the Fig. 7(b),

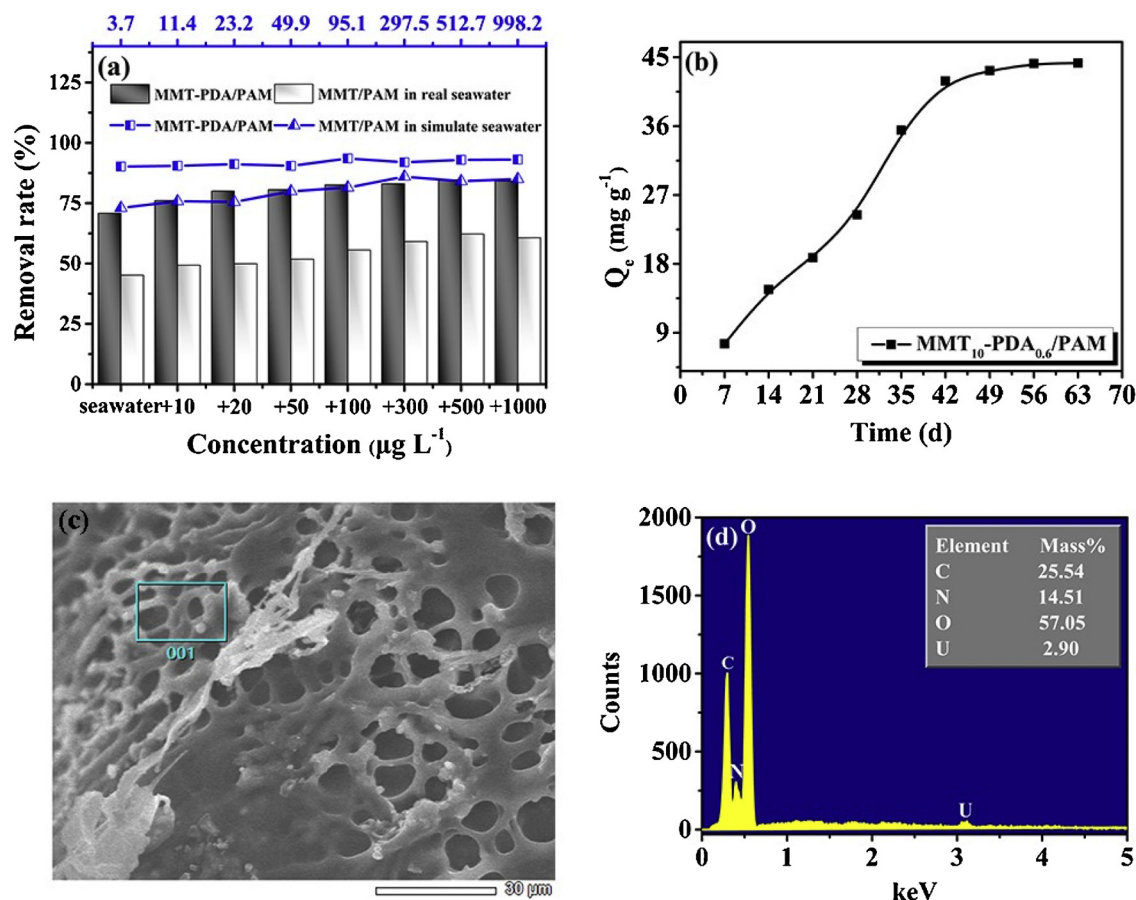


Fig. 7. (a) Removal rate of uranium by MMT₁₀/PAM hydrogel and MMT₁₀-PDA_{0.6}/PAM hydrogel from simulate seawater and real seawater, (b) the dynamic adsorption experiment with 400 L of circulated simulated seawater, (c) SEM and (d) EDS of the MMT₁₀-PDA_{0.6}/PAM hydrogel after dynamic adsorption experiment.

the amount of uranium uptake from simulated seawater reaches 44 mg g^{-1} after 63 days of contact between the adsorbent and simulated seawater in the flow column system. Significantly, the structure of the adsorbent did not change obviously after dynamic adsorption, suggesting adequate structural stability of the adsorbent materials (Fig. 7(c)). The EDS spectrum showed the presence of U, C, N and O on the surface of adsorbents (Fig. 7(d)), suggesting evidence of uranium adsorption after flow column experiment. To sum up, the above results indicated that MMT₁₀-PDA_{0.6}/PAM hydrogel shows great potential to be a new generation strategy for UES.

The desorption of uranium from the saturated adsorbents is essential from operating cost perspectives. 0.02 g MMT₁₀-PDA_{0.6}/PAM hydrogel after uranium adsorption can be regenerated by soaking in 0.05 mol L^{-1} $(\text{NH}_4)_2\text{CO}_3$ for 6 h in a state of shock, then the adsorbent is separated from the liquid by filtration and washed by deionized water three times before the next cycle adsorption experiment (Li et al., 2017b). The removal efficiency can still reach 81.63% after sixth cycle (Fig. S9). As shown in Fig. 8(a), after the second cycle, the MMT₁₀-PDA_{0.6}/PAM hydrogel exhibits a gradual decrease of the adsorption capacity. Nevertheless however, the recycled adsorbent still shows an uranium uptake capacity of 204.71 mg g^{-1} after the sixth adsorption cycle. The decrease of adsorption amount may due to the some active binding sites were occupied by the residual uranium. Furthermore, as shown in Fig. 8(b), the mechanical properties of the MMT₁₀-PDA_{0.6}/PAM hydrogel after six cycles of adsorption and desorption were also explored. From the representative tensile stress-strain curve for MMT₁₀-PDA_{0.6}/PAM hydrogels, it can be found that the stress decreased by 46% and the strain decreased by 20% after six cycles, which demonstrating its relatively good structural stability. Meanwhile, the FTIR spectra (Fig. S10) revealed that the stability of the MMT₁₀-PDA_{0.6}/PAM

hydrogel after six cycles.

3.5. Anti-biofouling assay

To evaluate the anti-biofouling effect of MMT₁₀-PDA_{0.6}, MMT₁₀/PAM and MMT₁₀-PDA_{0.6}/PAM, these as-prepared adsorbents were cultured with *Nitzschia* in algae-contained simulated seawater. The *Nitzschia* density of 1 st day and 8th day was shown in Fig. 9(a). For the above three adsorbents, the initial cell density was 1.5 ± 0.22 , 1.7 ± 0.12 , $1.6 \pm 0.16 \times 10^5$ number mL^{-1} , respectively. The cell density was observed again after 8 days contact, the cell density was 2.3 ± 0.18 , 2.4 ± 0.21 , 2.3 ± 0.26 , respectively. It was found that these adsorbents did not affect the growth of *Nitzschia*. This phenomenon proved that the biocompatibility of the prepared adsorbent. The attachment case of *Nitzschia* on the surface of three adsorbents were observed by optical and fluorescence microscopy. It can be seen from Fig. 9(c–d), when the *Nitzschia* settled for 1 day, the MMT₁₀-PDA_{0.6}/PAM hydrogel showed better anti-algae performance than the MMT₁₀-PDA_{0.6} and MMT₁₀/PAM hydrogel. After 8 days contact, MMT₁₀/PAM hydrogel were completely covered with *Nitzschia* cells. In contrast, fewer cells adhered to the surface of MMT₁₀-PDA_{0.6} and MMT₁₀-PDA_{0.6}/PAM hydrogel. In addition, further investigation on the adsorption behavior of three adsorbents in 1 mg L^{-1} algae-contained simulated seawater. As shown in Fig. 9(b), the results indicate that the adsorption processes reached equilibrium states earlier (after 4 d) and yielded higher uranium adsorption amount ($1986 \mu\text{g g}^{-1}$) in algae environments than MMT₁₀-PDA_{0.6} and MMT₁₀/PAM (5 d, $1524 \mu\text{g g}^{-1}$ and 6 d, $1077 \mu\text{g g}^{-1}$, respectively). The key factor that influences the anti-adhesion ability is the hydrophilicity of the adsorbent. As shown in Fig. S11(a), the coexistence of PDA and MMT significantly reduces the

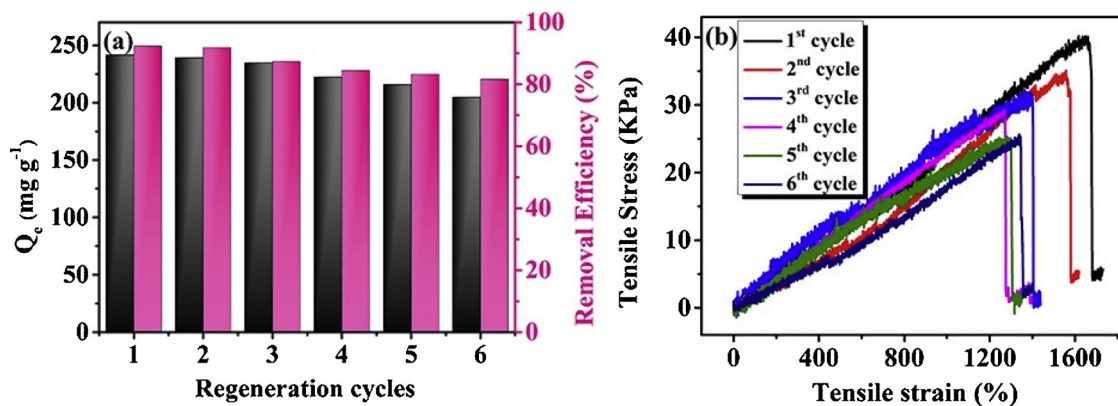


Fig. 8. (a) Uranium adsorption amount (left axis) and removal efficiency (right axis), (b) representative tensile stress-strain curves of MMT₁₀-PDA_{0.6}/PAM hydrogels after six adsorption-desorption cycles. $C_0 = 100 \text{ mg L}^{-1}$, amount of adsorbent = 0.02 g and the volume of uranium solution = 0.05 L.

contact angle of MMT₁₀-PDA_{0.6}/PAM. After contacting with water, the contact angle of MMT₁₀-PDA_{0.6}/PAM decreases rapidly and reaches the contact angle of 0° within 0.35 s after contact, indicating the high wettability of the material (Fig. S11(d)). The contact angle of the MMT₁₀-PDA_{0.6} and MMT₁₀/PAM decreases slowly than that of the MMT₁₀-PDA_{0.6}/PAM, and the contact angle of MMT₁₀-PDA_{0.6} and MMT₁₀/PAM are $9.67^\circ \pm 0.32^\circ$ (Fig. S11(b)) and $16.81^\circ \pm 0.28^\circ$ (Fig. S11(c)), when the contact angle of the MMT₁₀-PDA_{0.6}/PAM reaches 0°. This may be explained by the introduction of high hydrophilicity substance, namely, PDA and MMT into the hydrogel further increases the capacity of adsorbent to contain water. The water molecules from the

Nitzschia solution are preferentially adsorbed on the surface of the hydrogel, forming a layer of hydrated film on the surface, thereby weakening the interaction between Nitzschia and hydrogel, and finally achieving anti-adhesion (Jiang et al., 2014; Yang et al., 2017; Iqbal et al., 2012). Apart from this, the electrostatic repulsion between the negatively charged surface of the adsorbents and the negatively charged Nitzschia also result in this consequence.

3.6. Investigation of uranium binding interactions

The MMT₁₀-PDA_{0.6}/PAM hydrogel was further investigated their

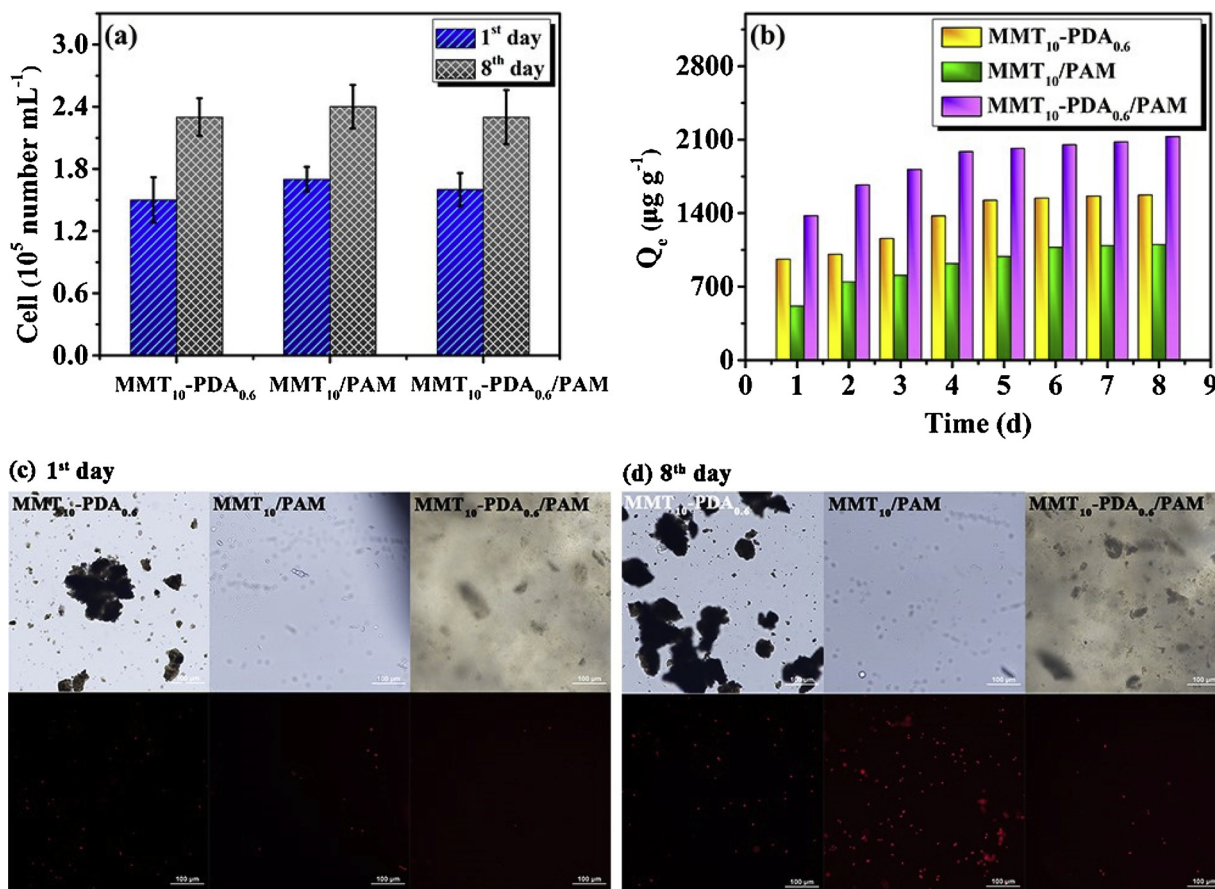


Fig. 9. (a) Nitzschia density of three adsorbents in solution on 1st day and 8th days, (b) adsorption of uranium in algae-contained simulated seawater. Conditions: pH = 8.3 ± 0.3 , T = 25 °C, $C_0 = 1 \text{ mg L}^{-1}$, $> 10^5$ number mL⁻¹ Nitzschia, optical and fluorescence microscopy images of Nitzschia adhering onto the three different adsorbents after (c) 1st day and (d) 8th days.

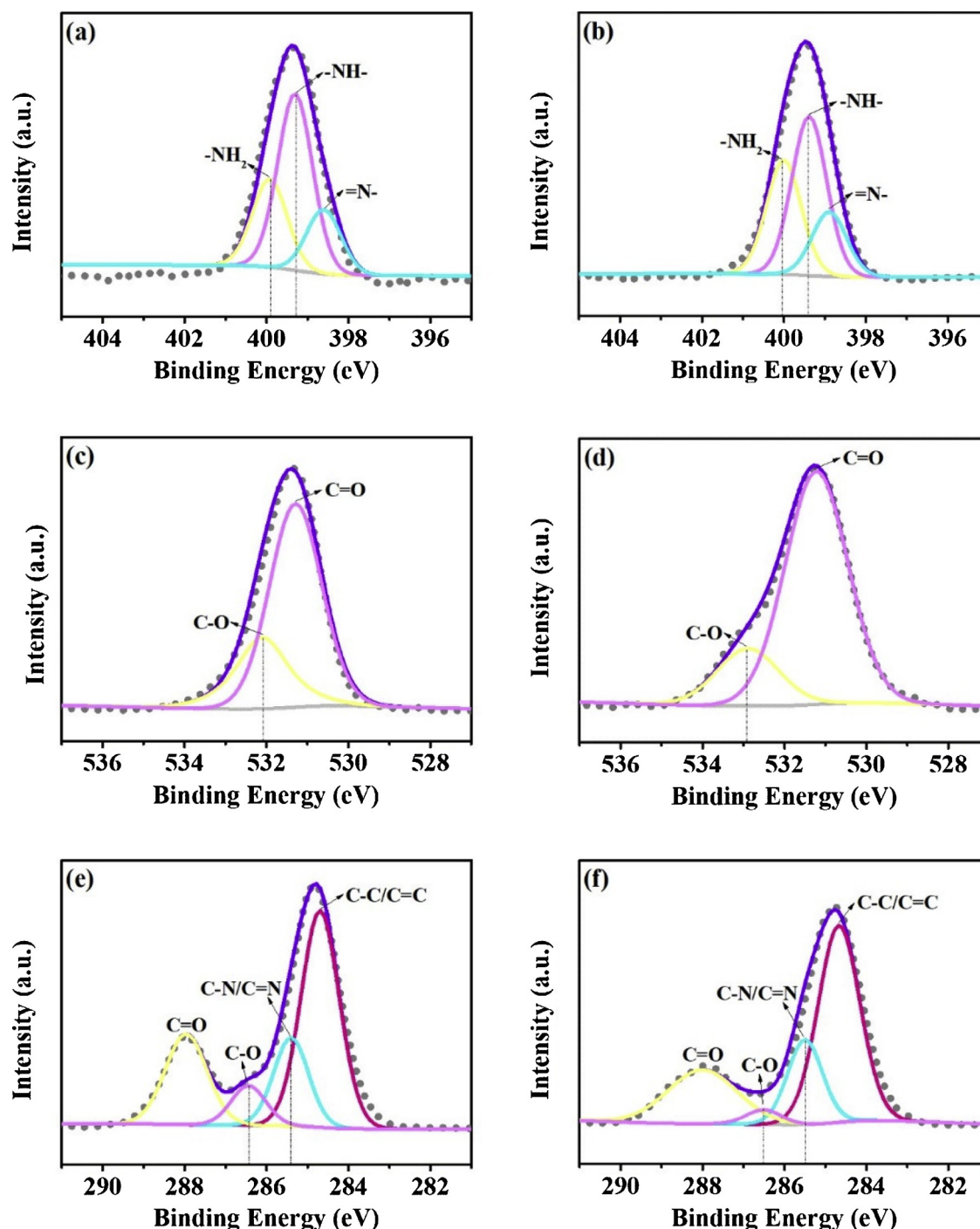


Fig. 10. Deconvolution of (a) N 1s, (c) O 1s, (e) C 1s XPS spectra of MMT₁₀-PDA_{0.6}/PAM and (b) N 1s, (d) O 1s, (f) C 1s XPS spectra of MMT₁₀-PDA_{0.6}/PAM-U.

remarkable binding interactions and uranium-binding environment via EDS mapping attached by SEM, FT-IR and XPS spectroscopy, aiming to supply more insights into the coordination interactions between the adsorbents and uranium. EDS mapping results showed a uniformly distribution of C, N, O and U species throughout the sample (Fig. S12(a–d)). FT-IR spectroscopy was carried out to detect the chemical binding of uranium in this adsorbents (Fig. S12(e)). The new band at $\sim 912\text{ cm}^{-1}$ for MMT₁₀-PDA_{0.6}/PAM-U hydrogel is assigned to the antisymmetric vibration of $\text{O}=\text{U}=\text{O}$ (Yuan et al., 2018b; Sun et al., 2018b). And the obvious upward trend shift of C–N stretching band (from 1596 to 1618 cm^{-1}) is ascribed to the metal-ligand binding compared MMT₁₀-PDA_{0.6}/PAM with MMT₁₀-PDA_{0.6}/PAM-U. To detect the mode of interaction between uranium and the adsorbent at the

molecular scale, XPS data of MMT₁₀-PDA_{0.6}/PAM hydrogel after uranium adsorption were obtained. The overall XPS surveys (Fig. S12(f)) represents the expected components of MMT₁₀-PDA_{0.6}/PAM hydrogel with the appearance of two U 4f signals, which verifies the existence of uranium. The U 4f can be resolved by a single contribution of uranium with U 4f_{5/2} ($\sim 392.1\text{ eV}$) and U 4f_{7/2} ($\sim 381.3\text{ eV}$), suggesting the valence state of uranium maintains unchanged during the adsorption (Fig. S12(g)).

The XPS of the N 1s spectrum of MMT₁₀-PDA_{0.6}/PAM separated into three peaks: -NH- (399.27 eV), -NH₂ (399.89 eV) and -N= (398.61 eV), which correspond to the cyclized nitrogen in the indole building blocks and the open-chain oligomers in PDA (Fig. 10(a)) (Wu et al., 2017). It can be noted that the -N= peak was not changed obviously after

binding to uranium, but the -NH- peak (399.40 eV) and -NH₂ peak (400.10 eV) were shifted to higher BE (Fig. 10(b)). This result implies that the -NH₂ species in the PDA aggregates and the -NH- species in the indole units might be participated in the coordination to uranium. The XPS analysis of the O 1s spectrum of MMT₁₀-PDA_{0.6}/PAM divided two signals, which corresponding C=O (531.30 eV) and C-O (532.15 eV) (Fig. 10(c)). After uranium adsorption, the C-O markedly shifted to higher BE of 532.91 eV, but the C=O signal almost unchanged (Fig. 10(d)). This suggests that the O atom that in catechol groups was largely involved in the complexation of uranium (Shi et al., 2014). And the XPS of the C 1s spectrum of MMT₁₀-PDA_{0.6}/PAM can be divided in four peaks: C=O (287.97 eV), C-O (286.36 eV), C=N/C-N (285.40 eV) and C=C/C-C (284.69 eV) (Fig. 10(e)) (Wu et al., 2016). After uranium adsorption, two signals were shifted to relatively higher BE region (C=N/C-N peak (285.53 eV) and C-O signal (286.61 eV)) (Fig. 10(f)), which was consistent with the change of the C=N/C-N species in N 1s spectra and the CO- species in O 1s spectra, respectively.

4. Conclusion

Overall, inspired by PDA chemistry and nanocomposite concept, a robust MMT-PDA/PAM hydrogel adsorbents with excellent adsorption capacity and anti-biofouling property via two-step was constructed in this work. The well-dispersion of polydopamine-intercalated montmorillonite nanosheet into polyacrylamide network improved the interfacial compatibility of nanocomposites, thereby shows satisfying mechanical behavior and stability. The three-dimensional porous building blocks provide high buried water content and numerous active sites coordinating to uranium. The maximum adsorption amount reached 617.28 mg g⁻¹ within 30 min, which is higher and faster than that of conventional hydrogel adsorbents. Meanwhile, the MMT₁₀-PDA_{0.6}/PAM showed good anti-adhesion activity, effectively preventing attachment of the *Nitzschia* after 8 days contact without inhibit the growth of *Nitzschia*. The adsorption of uranium was almost unaffected, which can reach 2130 µg g⁻¹ in algae-contained simulated seawater. Most importantly, in lab-scale continuous dynamic adsorption system, uranium uptake capacity of this MMT₁₀-PDA_{0.6}/PAM hydrogel can reach to 44 mg g⁻¹ in 63 days. Moreover, long service life of 6 adsorption-desorption cycles and acceptable mechanical strength indicate that this hydrogel can meets the real requirements for the application of UES. Our work thereby highlights a practical strategy that provides a new direction towards the development of efficient, and applicable hydrogel adsorbents for UES.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jhazmat.2019.120984>.

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